Optimus Primal — Gait Optimization

From Hand-Tuned Crawl to Learned Locomotion

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Problem

- **Slow iteration loop** — each hand-tuned gait takes minutes to test on hardware; no way to explore the full parameter space
- **Stride length hit a stability ceiling** — longer strides toppled the robot; couldn't find the right weight-shift timing by hand
- **Discrete hand-scripted phases don't generalize** — moving to simulation lets the robot optimize against *intrinsic* goals (keep the frame parallel to the ground) rather than fixed motor sequences

Digital Twin — URDF to MuJoCo

- Built from the same `optimus_primal.urdf` the real robot uses — same link geometry, joint limits, effort caps
- 8 actuated joints (4 hips + 4 knees) driven by **position servos** (k_p=2.5, k_v=0.05, initial guess; bench-measured later — see slide 13)
- Free joint on `base_link` — physics decides what happens; no kinematic cheating
- Floor: flat plane with friction μ=1.0 (rubber/TPU-sock approximation)
- **Caveat:** CAD wasn't modeled at the origin, so STL meshes render detached. Collision geometry is correct; visual hidden via `discardvisual`. Cosmetic only — physics unaffected.

Search Space — 105 Parameters

- Gait **structure is fixed**: 13 phases in a crawl cycle
  `start` $\rightarrow$ {`shift, swing, plant`} $\times$ {`FR, RL, FL, RR`}
  Optimizer tunes the **angles at each phase**, not the sequencing
- **104 joint-angle params:** 13 phases $\times$ 4 legs $\times$ 2 joints (hip + knee), in degrees
- **+1 phase-time param:** seconds between phases — controls gait cadence

Parameter	Lower Bound	Upper Bound
Hip angle	5°	55°
Knee angle	-100°	-25°
Phase time	0.15 s	1.2 s

- **Seeded from the hand-tuned gait** — generation 1 already walks
- **Sim-to-real is a parameter copy** — same degree convention as the robot's motor controller; winning JSON loads directly onto hardware

Algorithm — CMA-ES

Covariance Matrix Adaptation Evolution Strategy

Gradient-free, population-based optimizer — the standard for robot gait tuning.

Each generation:

1. Sample 48 candidate gaits from a multivariate Gaussian $\mathcal{N}(\mathbf{m}, \sigma^2\mathbf{C})$
2. Evaluate every candidate in MuJoCo — in parallel across CPU cores
3. Rank by reward, keep the top half
4. Update the distribution:
   - **Mean m** moves toward the winning region
   - **Covariance C** stretches along directions that produced good solutions
   - **Step size** σ grows or shrinks based on progress

Engineering: popsize $= 48$, $\sigma_0 = 5.0$, auto-restart on 15-gen stall, parallel evaluation ($\sim 7\times$ speedup on 8-core Mac), checkpoint every generation.

Reward Function

Shaped iteratively through failure.

Component	Purpose	Weight
Forward distance	Primary goal	$+100/\mathrm{m}$
Phases completed	Survival bonus	$+0.5/\mathrm{phase}$
Mean body height	Don't collapse	$+5$
Backward distance	Wrong direction	$-200/\mathrm{m}$
Mean pitch^2	Stay level (F/B)	-30
Mean roll^2	Stay level (L/R)	-40
Peak pitch/roll^2	No big momentary tilts	-8 each
Mean nose-down^2	**Don't flop forward**	$\mathbf{-200}$
Peak nose-down^2	**Not even once**	$\mathbf{-60}$
Mean pitch-rate^2	No floppy motions	-2

Hard terminations: body height $< 8\,\mathrm{cm}$, any tilt $> 30^\circ$, nose-down $> 27^\circ$.

Reward Function — Iteration Story

Each version was shaped by watching what the optimizer exploited.

v1 — Distance only Robot learned to collapse onto its shins and scoot forward. Technically moved. Not a gait.

v2 — Symmetric tilt penalty Tightened fall threshold from 45° to 30°. Better balance, but still toppled forward during swing phases.

v3 — Directional nose-down + pitch-rate Targets the specific forward-flop failure mode without penalizing backward lean (useful for weight shifting).

The reward function is as much the product of learning as the gait itself.

Results — Does the Starting Point Matter?

Three CMA-ES runs, same reward function, different seeds:

Hand-tuned gait Seeded from the manually-tuned `stance.py` crawl

Random Uniform random within bounds

Neutral standing Every phase pinned to the standing pose (no motion)

Takeaway: Seeding from a human-tuned gait gives a massive head start — generation 1 is already walking, so the optimizer spends its budget *refining* instead of *discovering*.

Results — Algorithm Comparison

Same reward function, same seed, different optimization algorithms:

CMA-ES Population-based, learns covariance structure of the search space

Differential Evolution Population-based, mutates by combining existing solutions

Random Search Uniform sampling baseline — "does optimization even help?"

Limitation — Open-Loop Control

CMA-ES found the best choreography. But it's still a choreography.

- Output is a **fixed lookup table**: 13 phases, 8 angles each, played back identically every stride
- No sensors, no feedback, no ability to react
- Works in sim (perfect flat floor) but struggles on real hardware

	CMA-ES (current)	What we need
Output	105 fixed numbers	A function that reacts
Sensors at runtime	None — blind playback	IMU, joint positions
Adapts to disturbance	No	Yes
What it learned	"The best script"	"How to walk"

Next — Reinforcement Learning

Close the loop: from choreography to skill.

RL trains a neural network (a "policy") that takes sensor readings *every timestep* and outputs motor commands:

$$\texttt{observe} \rightarrow \texttt{decide} \rightarrow \texttt{act} \rightarrow \texttt{observe} \rightarrow \texttt{decide} \rightarrow \texttt{act} \rightarrow \ldots$$

What we reuse:

- The MuJoCo digital twin — same sim
- The reward function — same objectives, already well-shaped
- The CMA-ES gait — baseline to beat, can seed the initial policy

What's new:

- Observation space: IMU (pitch, roll, $\dot{\theta}$) + 8 joint positions + 8 joint velocities
- Algorithm: PPO (Proximal Policy Optimization)
- Output: a neural network, not a JSON of fixed angles

CMA-ES found the best choreography for a perfect floor.
RL will learn to actually walk — adjusting every step based on what the robot feels.

PPO — Reward Engineering Journey (v5–v34)

30+ reward variants. Learning the failure modes was the work.

Era	What we tried	What broke
v5–v14	Distance + tilt + height + extension	"Reward-engineering ceiling" — combined ∼13
v15–v18	Stride-detection bonuses (**silent bug**)	Foot-name lookup returned –1; rewards never fire
v20	Post-fix recipe	Combined ∼42; new champion
v23–v26	Body-smoothness + foot-drift penalties	Body smoothness helped (N=2, noisy)
v27–v29	Survival bonus, fall-penalty schedules	Most regressed; high seed variance
v30	**Friction domain randomization**	*Best sim result, doesn't deploy*
v31	High kp (k_p=20) for deployable PD	Policy never learned to walk
v33–v34	X-config + low-friction + moderate steps	Same actuator-gap, 70–82% bang-bang

High-water marks: v20 (combined ∼42), v25_scratch_3M_b (44.7), v26_body_smooth_b (44.9).

Across 23 trials in 9 cohorts, within-cohort seed variance (±10–30) dominated between-cohort effects.

PPO learning curves — selected runs

Sim-to-Real — Friction Domain Randomization (v30)

Real surfaces aren't sim-default (μ=1.0). Trained three variants over $\mu \in [0.3, 1.0]$.

	μ=1.0	μ=0.7	μ=0.5	μ=0.3	$\Delta(1.0 \to 0.3)$
bodysmooth_dr	50.5 m	56.4 m	54.0 m	**53.9 m**	**+7%**
v20_dr	**61.2 m**	56.5 m	51.5 m	49.4 m	−19%
v6trig_dr	35.1 m	37.1 m	35.1 m	39.6 m	+13%

Finding: `bodysmooth_dr` is *flat across the friction range* — exactly what DR was supposed to produce. Pick for any low-friction deploy surface.

But — preparing this for hardware uncovered the **actuator gap**, a sim-to-real failure mode unrelated to friction.

The Actuator Gap — Why PPO Doesn't Deploy

Setup: Trying to load the v30 friction-DR champion onto hardware revealed something more fundamental than friction.

Diagnostic measurements:

1. **Bang-bang action saturation: 88%**
   Almost every commanded joint target is at the action-space LOW or HIGH.
2. **Joint qpos response oscillates hard**
   Under MuJoCo's PD (k_p=2.5, k_v=0.05), the joint slowly chases the bang-bang target. P95 $|\Delta q|$/step $= 17°$.
3. **Open-loop replay test: 65.8 m → 0.57 m**
   Take the joint trajectory the body actually followed, feed it as the env's target sequence in a fresh deterministic rollout. The body doesn't walk anymore.

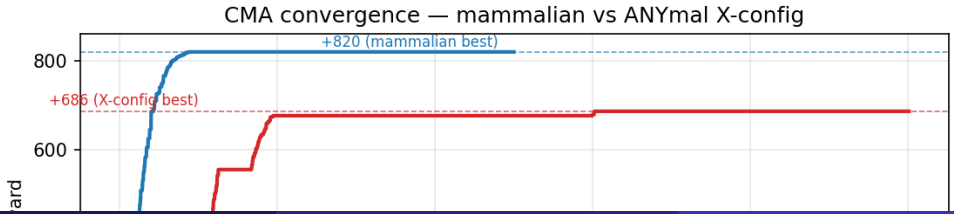
Interpretation: *Walking lives in the bang-bang action stream interacting with the PD, not in the resulting joint trajectory.* PPO discovered the underdamped PD acts as a low-pass filter; saturated commands at the right phase produce useful body motion. Same trick can't reproduce on the LX-16A's trajectory generator.

ANYmal X-Config — Geometry Experiment

Hypothesis: An X-stance (front knees forward, rear hips swept forward, all four feet under chassis) might be more stable to walk on than the mammalian default.

- Hand-posed the robot kinesthetically; captured per-leg servo angles
- Built `optimus_primal_xconfig.urdf` with rear hip + rear knee axes flipped
- Hardware `--xconfig` flag mirrors the same flip in `leg_abs`
- Visual sanity-check via `view_xconfig.py` — chassis stands level, four feet planted

X-config CMA: climbed cleanly to **best** $\approx +686$ by gen $\sim$300 ($\sim$84% of the mammalian gen-2500 winner at $\frac{1}{5}$ the compute).



Closing the Sim-to-Real Loop — Bench Calibration

Step-response on each LX-16A; fit MuJoCo PD to match.

Pi: `motor_step_response.py` steps each servo $\pm 10^\circ$, samples `get_physical_angle()` at 50 Hz, saves JSON.

Mac: `fit_pd_from_step_response.py` sweeps (k_p, k_v) for the best 2nd-order fit per servo.

Joint type	k_p	k_v	RMSE
Front hips	0.47	1.23	6.0°
Rear hips	1.14	1.92	5.4°
All knees	0.73	0.5–0.9	7.0°
Median	**0.73**	**0.87**	—

Two conclusions:

1. Original k_p=2.5 was *too stiff* — sim is a faster actuator than reality.
2. RMSE of 5–7° on a 10° step won't go away — **the LX-16A is a trajectory generator, not a PD**. No (k_p, k_v) matches.

The Fix — Velocity-Slew Actuator

Don't tune the gains. Change the actuator model.

Added `slew_rate_dps` parameter to `OptimusPrimalEnv.step()`: per-substep update to `data.ctrl` is clamped to `slew_rate` $\times$ `dt` of change toward the policy's action. Emulates the LX-16A's internal constant-velocity trajectory generator at the actuator level.

Verified: a bang-bang action stream now produces only $\sim 0.6^\circ$/env-step of joint motion at 125°/s slew rate (vs the unconstrained PD's $\pm 17^\circ$/step).

What this enables (untested at submission time):

- PPO action stream becomes the deployable trajectory — each action is a position target the LX-16A interpolates to over the 16 ms ctrl interval
- Bang-bang is impossible: the sim physically rate-limits the action stream before it ever reaches the PD
- Deployability gate: if `ppo_actuator_gap.py` reports saturation $< 30\%$ AND replay ratio $> 50\%$, the trace ports to hardware via `LX16A.move(angle, time=16)` calls

CMA gaits unaffected — they don't go through gym_env, and their slow phase times already

Where We Land — Submission Snapshot

What deploys to hardware today:

- **Mammalian CMA gait** (`best_gait.json`, +821 reward, gen 2500)
- **X-config CMA gait** (`cma_xconfig.json`, +680 reward, gen ∼300)
- Both load directly via `gait_controller.py --gait <file>`

What we learned that the slides don't fully convey:

- Reward shaping is a search problem within the search problem
- Seed variance dominates ablation effects until you commit to 5+ seeds per cell
- Sim-to-real is not one gap, it's a stack: friction, then actuator-model, then sensor model

What's next, ready but unrun:

- PPO with `--slew-rate 125 --kp 20` — the hypothesis is that the velocity-slew actuator removes the bang-bang exploit and makes PPO traces deployable
- Bench friction calibration (the same step-response idea, but for μ)

Optimizing the gait was easy compared to learning what we were optimizing for.